

how to magnetise and demagnetise steel

magnetise:

put inside a solenoid
connect solenoid to d.c. supply
switch on

demagnetise:

put inside a solenoid
connect to a.c. supply
withdraw rod from solenoid/slowly reduce current to zero
DO NOT switch off and take rod out directly, as then the magnet will have magnetic field which was created by the solenoid at the time of switching off.
OTHER METHODS:
pound with hammer
heat until red hot

experiment to determine magnetic field pattern

use of compass/suspended small magnet
observe needle/magnet on one field line
observe needle/magnet on another field line
mark on card
OR
sprinkle iron filings
tap card
direction/alignment of iron filings show field
use compass/suspended small magnet to show direction

how to increase speed of motor

more turns/several coils
iron core
increase current
stronger magnet
smaller air gap
curved poles(if not in the diagram)
poles closer
use split ring commutator(if not used already)

emf. def.

energy supplied/work done per unit charge
to drive charge round a complete circuit

how to do induction

-ve induced:
+ve charge attracts electrons which are -ve
electrons move towards +ve charge
unbalanced +ve charges left at opposite end

+ve induced:

-ve charge repels electrons which are -ve

electrons move away from -ve charge towards other end

unbalanced +ve charges left at opposite end

why is there a vacuum in cathode-ray tube

air would stop/weaken the electron beam OR electrons have no collisions;

purpose of stuff in cathode ray tube

heater: to heat the cathode; provide energy to electrons to jump off (work function)

cathode: to emit electrons/to undergo thermionic emission;

anode system: to attract and accelerate electrons;

to allow the electron beam to pass through to the screen/to focus the beam;

analogue and digital definitions

analogue = any reading possible/idea of continuous variation
of value of quantity

digital = idea of two discrete states only

ads of high voltage in supply cables

high voltage allows low/less current;

$$P = (I^2) * R ;$$

low/less/reduced heating effect, thus more efficient and less power lost;

why is it cheaper:

thinner cables needed

less metal used

less massive pylons

why use a split ring commutator

to reverse current every half turn;

so that rotation continues / current direction is kept same;

how a voltage in primary coil causes voltage in second coil

a.c./changing current in primary;

alternating magnetic flux/field/force in core;

field channeled to secondary coil by core OR core increases effect;

alternating magnetic field;

field cuts secondary coil;

changing flux linkage in secondary;

induces emf. in secondary;